

§11.5 Alternating Series

Homework : 3,7,13,17,23,27,35

Who : $\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 + (-a_2) + a_3 + (-a_4) + \dots$,

$a_n > 0$ (or $a_n < 0$) for all n . (交錯級數，正負相間) .

What :

Theorem1 :

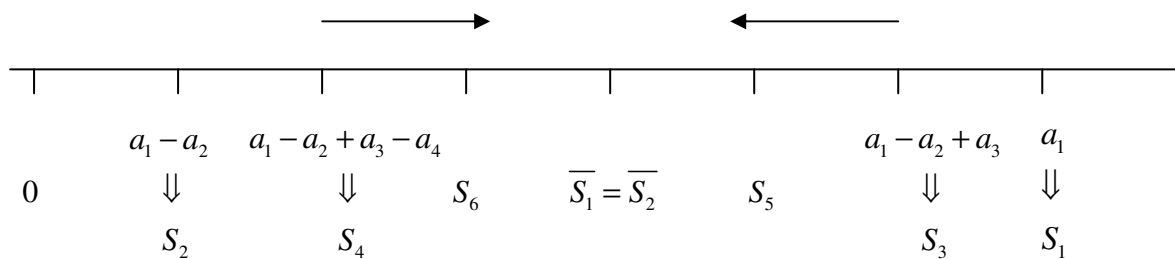
(i) a_n is decreasing.
(ii) $a_n \rightarrow 0$ as $n \rightarrow \infty$

(11.5-1)

$$\Rightarrow \sum_{n=1}^{\infty} (-1)^{n+1} a_n \text{ converges.}$$

Why :

(i) $a_1 > 0$



(ii) $S_{2n} \rightarrow \overline{S_2}$ and $S_{2n+1} \rightarrow \overline{S_1}$

$$\Rightarrow S_{2n+1} - S_{2n} = a_{2n+1} \rightarrow 0$$

$$\Rightarrow \overline{S_1} = \overline{S_2}.$$

Example 1 :

	Series	Conv./Div.
i.	$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$	Conv.
ii.	$\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$	Div. $\left(a_n \rightarrow \frac{3}{4} \neq 0 \right)$.
iii.	$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\ln n}{n}$	Conv.
iv.	$\sum_{n=1}^{\infty} (-1)^n \cos \frac{\pi}{n}$	Div. $(a_n \rightarrow 1)$.
v.	$\sum_{n=1}^{\infty} (-1)^n \sin \frac{\pi}{n}$	Conv.
vi.	$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{(\ln n)^p}{n}$	Conv.

$$\left(f(x) = \frac{(\ln x)^p}{x} \Rightarrow f'(x) = \frac{(\ln x)^{p-1} (p - \ln x)}{x^2} < 0, \text{ for large } x. \right)$$

Theorem2 : (Error Estimate) $\sum_{i=1}^{\infty} a_i$ satisfies (11.5-1).

$$|R_n| = |S - S_n| \leq a_{n+1}$$

Proof : $|R_n| = |S - S_n| \leq |S_{n+1} - S_n| \leq a_{n+1}$